



DevOps Copilot Application – Project Blog

Introduction

In today's fast-paced work environment, managing projects and tracking tasks efficiently has become very important for organizations. Many companies face challenges such as lack of task visibility, manual tracking processes, delayed communication, and difficulty in monitoring employee progress.

To solve these problems, we developed a **Task Management Application** using Microsoft Power Apps, Power Automate, and Microsoft Dataverse.

This application helps managers and employees streamline task assignment, tracking, approvals, and reporting through an automated workflow system.

Project Objective

The main objectives of this project were:

- To reduce manual task tracking
- To automate project workflows
- To improve communication between managers and employees
- To monitor task progress in real time
- To simplify the approval process
- To increase productivity and transparency

Technologies Used

1. Microsoft Dataverse

Dataverse was used as the backend database to securely store and manage all application data such as projects, tasks, employees, and worklogs. It also helped in maintaining relationships between different tables and ensured real-time data management.

2. Microsoft Power Apps (Model-Driven App)

The application was developed using a Model-Driven App in Power Apps. It provided a structured and professional user interface with built-in forms, views, dashboards, and role-based access control. It also helped in creating a scalable and easy-to-manage business application.

3. Power Automate

Power Automate was used to automate workflows and reduce manual work.

It handled:

- Email notifications
- Task status updates
- Approval workflows
- Automated alerts for managers and employees

This improved communication and increased process efficiency.

4. Business Process Flow (BPF)

Business Process Flow was implemented to guide users through each stage of the project and task lifecycle in a structured manner.

It helped in:

- Standardizing the workflow
- Tracking project stages
- Improving task management
- Ensuring process consistency

The BPF made the entire application workflow more organized and user-friendly.

Application Workflow

Step 1 – Project Creation

The manager first creates a new project and enters important details such as:

- Project ID
- Project Name
- Project Description

Step 2 – Task Assignment

After creating the project, the manager assigns tasks to employees.

Each task contains:

- Task ID
- Task Name
- Assigned To
- Project Name
- Milestone
- Priority
- Deadline

Once the task is assigned, the employee automatically receives a notification and email.

Step 3 – Employee Work Update

Employees can access their assigned tasks and:

- Update task progress
- Submit worklogs
- Change task status
- Mark tasks as completed

All updates are stored in real time in the database.

Step 4 – Approval Process

When an employee marks a task as completed, the manager receives a notification for review.

The manager can:

- Approve the task
- Reject the task
- Provide feedback or comments

If rejected, the employee can make corrections and resubmit the task.

Step 5 – Dashboard & Reporting

Interactive dashboards were created to display important KPIs such as:

- Total Projects
- Total Tasks
- Completed Tasks
- Pending Tasks
- Employee Performance
- Project Progress
- Hours Worked

These dashboards help managers make faster and better business decisions.

Database Structure

The application mainly uses the following tables:

Table Name	Purpose
Project_Details	Stores project-related information
Task_Details	Stores task assignment details
Work_Logs	Maintains employee work updates and task status

Key Features

- Real-time task tracking
- Automated notifications and emails
- Manager approval system
- Role-based access control
- Interactive dashboards
- Secure cloud database
- Workflow automation
- User-friendly interface

Benefits of the Application

For Managers

- Easy project monitoring
- Faster approvals
- Better productivity tracking
- Real-time reporting and insights

For Employees

- Clear task visibility
- Easy status updates
- Organized workflow
- Improved communication with managers

Challenges Faced During Development

During development, several challenges were faced:

- Managing data relationships
- Creating automated workflows
- Configuring role-based permissions
- Building dynamic email notifications

- Maintaining real-time updates

These challenges were solved through proper testing, workflow optimization, and database management.

Conclusion

The Task Management Application is a smart and efficient solution designed to simplify project handling and employee task tracking. By using Microsoft Power Apps, Power Automate, and Microsoft Dataverse, we developed an automated system that improves productivity, enhances transparency, and reduces manual effort.

In the future, this application can be enhanced with advanced features such as AI-based insights, performance analytics, and mobile push notifications.

